Allen Devaraj Augustin Ponraj

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Education

University of Colorado Boulder — Boulder, CO, USA
M.S. in Aerospace Engineering Sciences (GPA: 3.81/4.0) — *Aug 2025 – May 2027*

- Relevant Coursework: Linear Control Systems, Algorithmic Motion Planning, Statistical Estimation, Decision Making Under Uncertainty, Deep Reinforcement Learning, Small UAS Guidance, Navigation and Control

Madras Institute of Technology, Anna University — Chennai, Tamil Nadu, India
B.E. in Aeronautical Engineering (GPA: 3.51/4.0) — *Nov 2021 – May 2025*

- Relevant Coursework: Avionics Systems, Rocket Propulsion, Microcontroller Programming

Publications

- **A. Devaraj***, W. Xie*, N. Correll. "Leaning Over to Reach Far-Away Objects." IROS 2026 Workshop on Perception and Decision Making for Athletic Humanoid Robotics, Pittsburgh, PA. Poster, lightning talk and under review for ICRA 2027. Sep 2026
- Z. Allen*, M. Conway*, L. Antieau, **A. Ponraj**, N. Correll. "RAPID: Robotic Agentic Platform for Intelligent Electric Vehicle Disassembly." IEEE International Conference on Automation Science and Engineering (CASE). arXiv:2603.18520 2026
- M. Conway*, W. Xie, **A. Devaraj**, Y. Zhang, N. Pudasaini, M. Feit, A. Abid, Z. Allen, C. Liu, X. Tan, J. Lavering, J. Chen, L. Antieau, A. Von Pischke, A. Roncone, Z. Sunberg, N. Correll. "GOLEM: Modular Humanoid Autonomy Towards Electric Vehicle Battery Disassembly." arXiv preprint. arXiv:2608.21550 2026

Experience

Graduate Research Assistant — Boulder, CO, USA
Correll Lab – Intelligent Robotics Laboratory — *Jan 2026 – Present*

- **Humanoid whole-body control on the Unitree H1-2**
  - Built a sampling-based whole-body model predictive controller (MuJoCo MPC, CEM) for supported reaching on a real Unitree H1-2 humanoid: the robot braces a forearm on a table to extend its reach by about 50 cm beyond unbraced standing, holds the loaded posture for over four minutes, and recovers, using only onboard proprioception fused with AprilTag fiducials, no motion capture.
  - Achieved 89% target reach and 81% full reach-and-recovery over 54 hardware trials across six targets and three table heights, with end-effector accuracy of 2.7 ± 0.9 cm, replanning at 34 Hz on a laptop.
  - Built an LLM agent skill (Claude Code) for the H1-2 digital twin that replays run logs into a timed event map (contacts, actuator saturation, joint-limit proximity, fall onset), audits planner-versus-twin model parity, and autotunes MPC cost weights scored on the twin, used for failure diagnosis and weight tuning throughout the bracing work.
- **Robotic EV battery disassembly (ARPA-E, DoE)**
  - Developed the constrained motion planning pipeline with analytical IK seed ranking and cost-normalized configuration selection for a 7-DOF industrial manipulator on a gantry, the manipulation layer of RAPID, where SmolAgents LLM agents call ROS tools to sequence the disassembly.

ADCS Engineer, SWARM-EX CubeSat (NSF) — Boulder, CO, USA
Colorado Center for Astrodynamics Research — *Sept 2025 – Present*

- Co-lead the ADCS subsystem (Blue Canyon XACT-15): verified its command and telemetry set through the HydraGUI ground-station software and built the XACT telemetry pages and packet parser used for bench testing.
- Routed NovAtel OEM719 GPS position and time into the XACT in the CDH flight code, and set up the first CDH-to-SWAG GNC interface for formation control.

Project Intern – ADCS Testbed for CubeSats — Trivandrum, Kerala, India
SSPACE Lab, Indian Institute of Space Science and Technology — *Jun 2024 – Jul 2024*

- Developed and implemented a one degree-of-freedom control algorithm for a single reaction wheel on a FloatSat with air bearings to test CubeSat ADCS systems.
- Achieved precise wireless Z-axis stabilization and control for the FloatSat platform.

Projects

SLAM with DeepRacers Feb 2026 – May 2026

- Implemented SLAM for an autonomous rover navigating underground corridors using LiDAR, IMU, and odometry without wheel encoders.
- Developed a global A* path planner integrated with a Pure Pursuit geometric controller for autonomous navigation.
- Built a vision-based line-following pipeline using an Intel RealSense camera and image segmentation techniques.

Belief-Augmented RL for Pick-and-Place under Uncertainty Jan 2026 – Apr 2026

- Evaluated standard PPO and belief-based PPO reinforcement learning approaches under varying uncertainty and noise conditions.
- Simulated uncertainties including distance-dependent sensor noise, occlusion-induced observation dropout, and workspace drift noise.
- Developed a simulated manipulation environment where an SO101 robotic arm autonomously picks a red LEGO block from mixed blue blocks without a vision pipeline.

Dual Manipulators Performing Insertion Task using TAMP Aug 2025 – Dec 2025

- Performed a coordinated peg-in-hole insertion task using two simulated Franka manipulators through Task and Motion Planning (TAMP): symbolic task planner, geometric reasoner, and MoveIt/OMPL motion planner with a GUI for dual-arm control and failure reporting.

Methods of Stabilization in Ground Effect Aircrafts Aug 2024 – May 2025

- Engineered stabilization methods for Wing-In-Ground Effect Aircrafts via CFD-based SOGA optimization of tail geometry, achieving a 9.468% increase in lift curve slope.
- Designed a feedback control system for pitch damping to improve longitudinal aircraft stability.
- Contributed to the design and fabrication of a remote-controlled aircraft for testing.

AeroTHON 2024 – Team Classified Mar 2024 – Nov 2024

- Led a team in an SAEINDIA competition to design and build a quadcopter with payload delivery, image processing, and autonomous capabilities.
- Handled systems integration, flight software development, and onboard image-processing pipelines.
- Trained a YOLOv8 model for live onboard object classification and implemented autonomous Boustrophedon path planning using ROS2 on hardware.

CanSat 2024 – Team Canopy Oct 2023 – Mar 2024

- Led the conceptualization of a Can Satellite capable of ejecting a heat shield and deploying a parachute for safe payload landing from 725 m altitude.
- Managed CAD modeling, flight software programming, and ground station software development.

3D Trajectory Plotting of a Sounding Rocket Aug 2023 – Nov 2023

- Led a 4-member team in the design and pre-flight testing of a sounding rocket equipped with a custom avionics bay for telemetry acquisition.
- Engineered a parachute ejection mechanism and contributed to flight software and PCB prototyping.
- Developed a MATLAB application to analyze IMU telemetry and generate 3D trajectory plots of the sounding rocket.

Achievements

- Received the "Giri Brothers Award" for the Best Senior Project at MIT Chennai. 2025
- Secured 12th rank internationally as Team Canopy in the Design Phase of CanSat 2024. 2024
- Secured 11th rank nationally as Team Classified in AeroTHON 2024. 2024
- Received the "Amita Award" for securing the highest score in Python Programming during sophomore year at MIT Chennai. 2022

Certifications

- Autodesk Fusion 360 Simulation on Coursera (Jul 2023) – Certificate ID: 5W8JUDUJGVSF
- Drone Simulation in Simulink on Udemy (Jun 2024) – Certificate ID: UC-e80f3998-5339-407d-a5e0-be257059fa01

Service

- Peer Reviewer, IEEE Transactions on Robotics (T-RO) 2026
- Peer Reviewer, IROS 2026 Workshop on Perception and Decision Making for Athletic Humanoid Robotics 2026

Workshops and Seminars

- Presented a poster and lightning talk at the IROS 2026 Workshop on Perception and Decision Making for Athletic Humanoid Robotics, Pittsburgh, PA Sep 2026
- Represented CU Boulder at the ARPA-E Energy Innovation Summit, presenting an ARPA-E-funded robotics research project to industry and government stakeholders, San Diego, CA Apr 2026
- National-level technical seminar on "Emerging Technologies in Aerospace" at MIT, Anna University May 2024
- Flight Training Laboratory at the Indian Institute of Technology Kanpur, India Feb 2024
- STK Software Training conducted by SS Technologies, Bangalore, India Oct 2023
- Elements of Aeronautics with Robotics at Warsaw University of Technology, Poland Jul 2023

Technical Skills

CAD Design: Autodesk Fusion 360, CATIA, SolidWorks
PCB Design: Autodesk Eagle, Proteus
Programming Languages: Python, C++, MATLAB
Microcontrollers & Embedded Systems: Arduino IDE, Raspberry Pi, Pixhawk
Robotics Frameworks & Simulation: ROS 2, MuJoCo, MuJoCo MPC, Gazebo, PyTorch
Simulation: Autodesk CFD
Data Analysis: Mission Planner, QGroundControl
Manufacturing Processes: Soldering, 3D Printing

Portfolio

allendevaraj.github.io